Bartosz Bojda

UNITY DEVELOPER · VR & MULTIPLAYER

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PROFILE

Unity developer with **a decade of experience shipping VR titles** across Meta Quest, Valve Index, WMR, desktop and mobile — including **Elven Assassin**, one of the top-grossing arcade titles on the VR market. Nine shipped titles, with a focus on multiplayer netcode, performance optimization against strict VR frame budgets, and the gameplay systems built on top of both. Four years in-office and six remote.

SKILLS

ENGINE & ARCHITECTURE	C#, Unity, Design Patterns, SOLID Principles, Zenject / VContainer (Dependency Injection), R3 (Reactive Programming), Unity DOTS / ECS, Unit Testing
VR & GRAPHICS	OpenXR, VR Development (Meta Quest, Valve Index, Pico, WMR), Meta Platform SDK, Hurricane VR, FinalIK, Shader Development & Debugging, Logic & Graphics Optimization, Unity Addressables, UI Implementation
MULTIPLAYER & BACKEND	Photon Unity Networking (PUN), Netcode & State Synchronization, Cloud-Hosted Game Services (AWS), REST APIs, Backend Database Integration
AUDIO & INTEGRATION	FMOD Audio Integration, Platform SDKs (Steam, Meta), Cross-Platform Porting (VR, Desktop, Mobile), PaintIn3D
PIPELINE & PROCESS	Git, Jenkins (Automated Builds / CI-CD), Jira, Agile / Scrum, Code Review, Internal Coding Standards
AI-ASSISTED DEVELOPMENT	Agentic Development, Unity MCP (Model Context Protocol), Claude Code

EXPERIENCE

Unity Programmer
03/2025 – 01/2026

VR Factory Games SA / Remote

- Contributed across the full development lifecycle and shipped release of **Battlecrafter VR**, including subsequent post-launch updates.
- Maintained and developed post-launch updates for **Workshop Simulator VR**, focusing on feature enhancements and bug fixing.
- Developed primarily against **OpenXR**, integrating vendor SDKs where platform-specific features required it.
- Ported Battlecrafter VR to flat-screen PC, adapting VR-specific controls and mechanics for standard desktop systems.
- Implemented diverse gameplay mechanics, responsive user interfaces, and core technical systems.
- Optimized game logic and graphics to meet strict performance requirements and maintain high framerates in VR.
- Integrated complex audio systems using FMOD to enhance the game's soundscape and immersion.
- Implemented advanced dynamic texture painting mechanics using the PaintIn3D package.
- Modified custom shaders to resolve rendering issues and optimize visual performance.
- Integrated Meta APIs to support platform-specific features and functionality.
- Maintained high standards of code quality through clean coding practices and design patterns.
- Collaborated within the development pipeline using Git, Jira for task tracking, and Jenkins for automated builds.

Unity Developer
05/2016 – 11/2024

Wenkly Studio / Gliwice & Remote

- Developed and shipped numerous VR and mobile titles, including **Elven Assassin**, **Survival Nation**, **Astro Hunters**, **Private Property**, **Battle of Kings**, and Google Cardboard titles (Last Order: Blackwood Project, Skydive VR).

- Implemented core gameplay mechanics, advanced AI systems, responsive UI, and robust networking solutions across multiple projects.
- Engineered multiplayer systems and netcode using Photon Networking to deliver seamless online experiences.
- Designed and built an automated tournament system: a headless host application running unattended on a cloud machine, launching tournament stages, assigning players to matches and reporting results to a backend over REST — defined the backend requirements and API contract with the infrastructure provider.
- Optimized game logic and graphics to ensure stable performance and high framerates across target platforms.
- Ported titles across diverse platforms including Valve, Meta and WMR headsets, desktop and mobile, ensuring cross-platform compatibility.
- Integrated platform-specific APIs from Steam, Meta, and other major distributors.
- Refactored legacy codebases to significantly improve code quality, maintainability, and architectural scalability.
- Established and introduced internal programming standards to elevate team-wide code consistency.
- Participated actively in intensive code reviews, fostering a culture of high-quality development and knowledge sharing.
- Collaborated in an Agile/Scrum framework, managing tasks and workflows via Jira and Git.

SELECTED PROJECTS

Elven Assassin

Wenkly Studio

A room-scale VR archery game: defend a town from waves of orcs with a bow, solo or in four-player co-op. Recognized as one of the top-grossing arcade titles on the VR market.

- Built the tournament host application and the game client's REST integration, and defined the backend requirements and API contract with the infrastructure provider — tournaments ran start to finish on a cloud machine without anyone from the studio present.
- Engineered a flexible wave system allowing dynamic difficulty adjustment and game-balancing tuning.
- Designed and built an extensible spell system focused on modularity and long-term reusability.
- Optimized enemy pathfinding to handle complex movement and maximize runtime performance.
- Developed cross-platform deployment tools to streamline project adaptation across target hardware.

Battlecrafter VR

VR Factory Games

A Meta Quest restoration simulator set in a Martian war museum. Players rebuild historical weapons and armor using precision tools; restored pieces are installed into multimedia domes that recreate iconic battles.

- Engineered core restoration and assembly mechanics, including dynamic tool interactions that modify object graphics and handle full part assembly and disassembly.
- Developed physics-based object manipulation within a specialized "antigravity device" mechanic.
- Developed an in-game store system using currency earned from restoration contracts to purchase durable upgrades (tools, attachments) and consumables.
- Built a flexible voiceover/lector system to handle dynamic narration and audio cues.
- Designed an intuitive "How to Play" tutorial system to seamlessly onboard new players.
- Integrated platform achievements to enhance player engagement and completion loops.
- Ported core VR tool mechanics to flat-screen PC, ensuring cross-platform gameplay consistency.

Survival Nation

Wenkly Studio

An open-world online VR survival RPG with solo, PvE and PvP modes: zombie combat, scavenging and crafting, quest lines, and skill-tree progression.

- Integrated and configured the Hurricane VR plugin, significantly reducing development time while elevating the quality of VR interactions.
- Engineered AI behaviors for range-attacking enemies, implementing customized decision-making and combat logic.
- Developed and optimized a comprehensive player needs system to manage survival mechanics.
- Implemented asynchronous I/O using Addressables for dynamic font asset loading and unloading, optimizing memory management.

Astro Hunters
Wenkly Studio

A PvPvE extraction shooter set in space: scavenge alien planets for resources, craft and upgrade weapons aboard your mothership, then extract before rival players take the loot.

- Architected and implemented an advanced, interactive extraction mechanic as a core gameplay system.
- Developed a physics-based tablet UI interaction system to enhance player immersion and spatial usability.
- Engineered intelligent AI systems for firearm-equipped enemies, featuring sophisticated combat and tactical behaviors.
- Designed and programmed fluid traversal mechanics, including responsive player double-jump and hovering systems.

Private Property
Wenkly Studio

A co-op VR zombie shooter set in 1890s Montana, where teams defend their property from sprinting undead with revolvers, shotguns, rifles and a bow.

- Engineered comprehensive core gunplay systems, including modular shooting mechanics, reloading logic, and procedural weapon recoil.
- Implemented high-performance projectile tracking using frame-by-frame raycasting to handle dynamic physics and accurately trigger hit visual and audio effects.
- Developed and integrated an interactive stationary Gatling gun mechanic designed for player use.

Workshop Simulator VR
VR Factory Games

A highly immersive VR restoration simulator focused on detailed mechanical disassembly, tool interactions, and physical item repairs.

- Engineered contextual UX guides for precision tool interactions, implementing visual aids and micro-feedback loops for mechanical screw manipulation.
- Developed an in-game patch notes system, creating a dynamic UI component to seamlessly display update logs and version details to users.

Battle of Kings
Wenkly Studio

Tower-defense strategy game released on VR headsets and flat-screen PC: hold your fortifications under siege while developing an economy and upgrading your army, across single-player campaign and multiplayer PvP.

- Engineered the core gameplay logic and state management for both the single-player CPU campaign and multiplayer PvP modes.
- Developed a modular castle expansion system, managing progression structures, resource costs, and base-building mechanics.
- Designed and implemented a responsive camera movement system, ensuring smooth navigation and optimal tactical overview for the player.

EDUCATION

BSc Computer Science
Silesian University of Technology, Gliwice / 2013 – 2017

LANGUAGES

Polish Native

English C1